

Docker?

봉정근

Docker engine install (repo add)



```
1 # Add Docker's official GPG key:
2 sudo apt-get update
3 sudo apt-get install ca-certificates curl
4 sudo install -m 0755 -d /etc/apt/keyrings
5 sudo curl -fsSL https://download.docker.com/linux/ubuntu/gpg -o /etc/apt/keyrings/docker.asc
6 sudo chmod a+r /etc/apt/keyrings/docker.asc
7
8 # Add the repository to Apt sources:
9 echo \
10 "deb [arch=$(dpkg --print-architecture) signed-by=/etc/apt/keyrings/docker.asc] https://download.docker.com/linux/ubuntu \
11 $(. /etc/os-release && echo "${UBUNTU_CODENAME:-$VERSION_CODENAME}") stable" | \
12 sudo tee /etc/apt/sources.list.d/docker.list > /dev/null
13 sudo apt-get update
```

Package install



```
1 sudo apt-get install docker-ce docker-ce-cli containerd.io docker-buildx-plugin docker-compose-plugin
```



```
1 sudo groupadd docker
2
3 sudo usermod -aG docker $USER
```

설정 후 재 로그인 필수

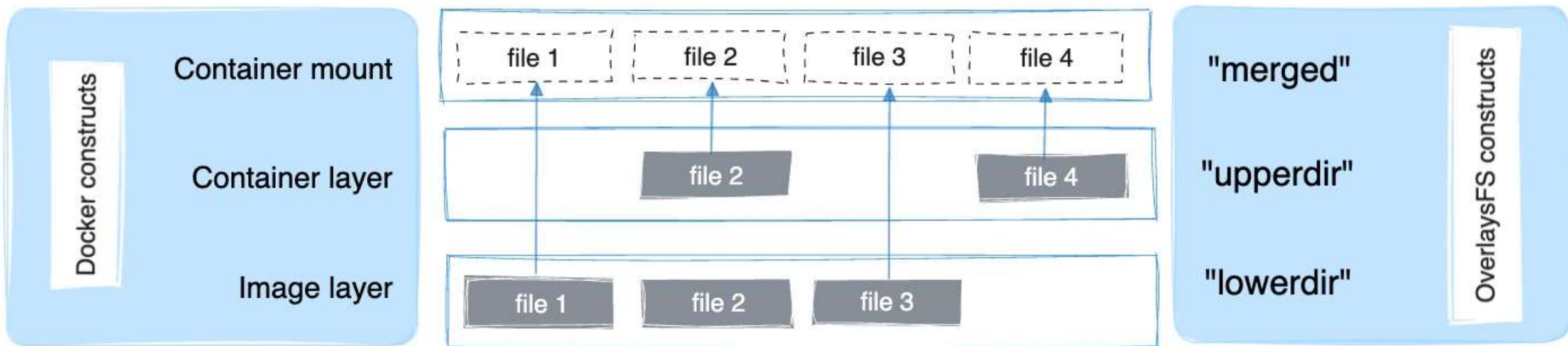
Docker command



```
1  ```\n2  docker (container) run - 컨테이너 실행\n3  docker (container) stop - 컨테이너 중지\n4  docker (container) rm - 컨테이너 삭제\n5  docker (container) exec - 컨테이너 접근\n6  docker (image) build - 이미지 빌드\n7  docker (image) pull - 이미지 가져오기 (default : docker hub)\n8  docker (image) push - 이미지 업로드 (default : docker hub)\n9  docker (image) tag - 이미지 태그 생성\n10  ```\n
```

OverlayFS (overlay2)

- 두개 이상의 디렉토리를 계층화하여 하나의 디렉토리로 표시하는 기술



OverlayFS (overlay2)

- 저장 효율이 좋음
- 단일 페이지 캐시로 메모리 효율이 좋음
- 상대적으로 쓰기 작업이 느림(CoW 방식과 동일)

Limitations on OverlayFS compatibility

To summarize the OverlayFS's aspect which is incompatible with other filesystems:

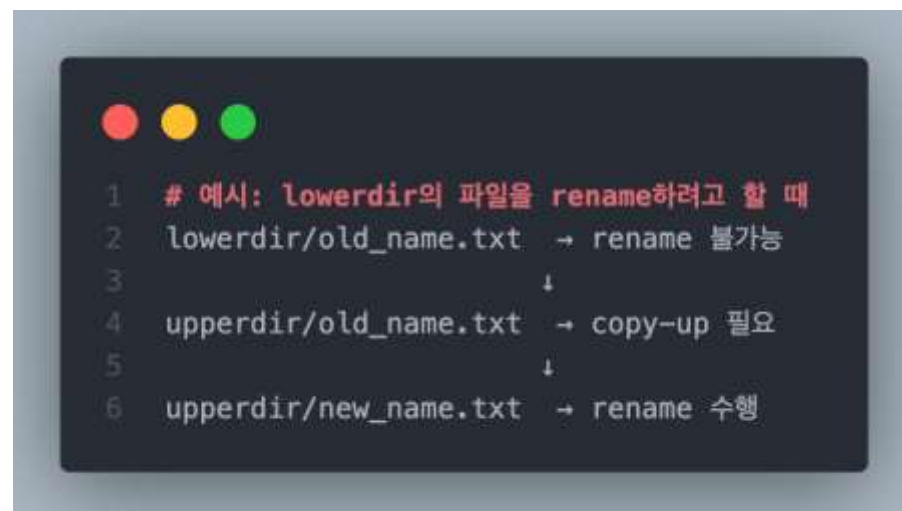
`open(2)`

OverlayFS only implements a subset of the POSIX standards. This can result in certain OverlayFS operations breaking POSIX standards. One such operation is the copy-up operation. Suppose that your application calls `fd1=open("foo", O_RDONLY)` and then `fd2=open("foo", O_RDWR)`. In this case, your application expects `fd1` and `fd2` to refer to the same file. However, due to a copy-up operation that occurs after the second calling to `open(2)`, the descriptors refer to different files. The `fd1` continues to reference the file in the image (`lowerdir`) and the `fd2` references the file in the container (`upperdir`). A workaround for this is to `touch` the files which causes the copy-up operation to happen. All subsequent `open(2)` operations regardless of read-only or read-write access mode reference the file in the container (`upperdir`).

`yum` is known to be affected unless the `yum-plugin-ovl` package is installed. If the `yum-plugin-ovl` package is not available in your distribution such as RHEL/CentOS prior to 6.8 or 7.2, you may need to run `touch /var/lib/rpm/*` before running `yum install`. This package implements the `touch` workaround referenced above for `yum`.

`rename(2)`

OverlayFS does not fully support the `rename(2)` system call. Your application needs to detect its failure and fall back to a "copy and unlink" strategy.



EXDEV

oldpath and *newpath* are not on the same mounted file system. (Linux permits a file system to be mounted at multiple points, but `rename()` does not work across different mount points, even if the same file system is mounted on both.)

```
3- "LowerDir": "/mnt/nvme/990_PR0_4TB/docker/overlay2/kn9r0hjyrzu76cw54md9livr/diff:
4- /mnt/nvme/990_PR0_4TB/docker/overlay2/49rjluvta082x71zpyvvnj01zh/diff:
5- /mnt/nvme/990_PR0_4TB/docker/overlay2/tm4k9sqxxshmu6n5b33fq1up/diff:
6- /mnt/nvme/990_PR0_4TB/docker/overlay2/vny8kwjligdojx3kng5jwk9il/diff:
7- /mnt/nvme/990_PR0_4TB/docker/overlay2/t6wfnpfoo1yv7ocuhm9f7axum/diff:
8- /mnt/nvme/990_PR0_4TB/docker/overlay2/gzppnmckiq1pal7q836k9h4xz/diff:
9- /mnt/nvme/990_PR0_4TB/docker/overlay2/neqalsluljglzmx4a1iobjvn/diff:
10- /mnt/nvme/990_PR0_4TB/docker/overlay2/5zangfotyf9uyit7y5mt13dea/diff:
11- /mnt/nvme/990_PR0_4TB/docker/overlay2/d3qc25pr2etyfo7lzny761l8l/diff:
12- /mnt/nvme/990_PR0_4TB/docker/overlay2/3ymoas2002ia0qv4ix0wa0p5o/diff:
13- /mnt/nvme/990_PR0_4TB/docker/overlay2/roy4dl6iuzbxjfi7t4s6pkcqt/diff:
14- /mnt/nvme/990_PR0_4TB/docker/overlay2/l9li282rar9graqv55gw7o127/diff:
15- /mnt/nvme/990_PR0_4TB/docker/overlay2/a9yxjwwdq1dncqwt94z1lt8z/diff:
16- /mnt/nvme/990_PR0_4TB/docker/overlay2/6mqskqnye4b9g54daee4sr4l4/diff:
17- /mnt/nvme/990_PR0_4TB/docker/overlay2/f6y7lcyfkval9qb79cmp0r993/diff:
18- /mnt/nvme/990_PR0_4TB/docker/overlay2/kdbdeuudp5dshlc450ysuv5ns/diff:
19- /mnt/nvme/990_PR0_4TB/docker/overlay2/l27xq8i4yktlaa3j3n3r2aqc4/diff:
20- /mnt/nvme/990_PR0_4TB/docker/overlay2/m38ag01icnhxkzavz8vip7jmj/diff:
21- /mnt/nvme/990_PR0_4TB/docker/overlay2/
22- 72de23bd713fa6cc6f46c091d65a06653d619816c9ca1b2b9118d6f243dff50d/diff:
23- /mnt/nvme/990_PR0_4TB/docker/overlay2/
24- 34fc565b2b05ef51f167ff5ac9f7c5d8e12360bcd081d07b186448c75d0ab64/diff:
25- /mnt/nvme/990_PR0_4TB/docker/overlay2/
26- 4897143df74cda136c5323c6513f4d1dad8668f87a9ce00ca4b612b56d45a010/diff:
27- /mnt/nvme/990_PR0_4TB/docker/overlay2/
28- 16e484f940327af75876efaf2035cc2adf54f795c3defea50389c5809d888f5e/diff",
29- "MergedDir": "/mnt/nvme/990_PR0_4TB/docker/overlay2/ywqxbfwx5ynu97cpsa97j507u/merged",
30- "UpperDir": "/mnt/nvme/990_PR0_4TB/docker/overlay2/ywqxbfwx5ynu97cpsa97j507u/diff",
31- "WorkDir": "/mnt/nvme/990_PR0_4TB/docker/overlay2/ywqxbfwx5ynu97cpsa97j507u/work"
32- },
33- "Name": "overlay2"
34- },
35- "RootFS": {
36- "Type": "layers",
37- "Layers": [
38- "sha256:418dcc7d85a63a6aa574439840f7a6fa6fd2321b3e2394568a317735e867d35",
39- "sha256:689195d1c9adf3fad82fbd3e2d279fa1bee001c73287e7e9ee2ed7a1881f2424",
40- "sha256:fddb0c9d0c8fa53f1466bdcc0baf00ad6ab1f3b5bcc5bdd152d3ce98f88ca16",
41- "sha256:00d0857b2a165e6b4dd62bd0d23f87e4e63a809a02b77333fe4351adadd52da7",
42- "sha256:754644eead3e70d8ddf4256935b64628cda167281e4425db8c29285fc83ed396",
43- "sha256:da963dab6af78b7e915a4c5f29966b14852df00bf99bec38c8e3a7400d83b9a9",
44- "sha256:f889c69583d2f7a7f1b01c0e15e023568cfd9a50e906eac9033ebb3d081dce99",
45- "sha256:36eb4331f20150d823f52aa204fe067201ba08b07ed57700400842bb1ade2045",
46- "sha256:2afdf1bef93cd49f4bc78cd6ddd5bb1bf9c2d86e060acbf117b86eb36ad5142c",
47- "sha256:eaed4d01f1befcd67ef089e2ef62f18cbaf7dca9898585ccb4b6968d379514771",
48- "sha256:bbdf35898477a1b8492e72abfe40835b46995b69fb0552bdaf77d1f500024f25",
49- "sha256:94eb5fa4282b81ab43f220baf41a84bca74b4c8389316792a4f20bba8fb7b10",
50- "sha256:5f70b1f18a086007016e948b04aed3b82103a36bea41755b6cddf10ace3c6ef",
51- "sha256:0bb3248bd9fe9a704ec8644eee9f4e6560c31d4b71faf6f7c10472efc478b18a",
52- "sha256:f01a81cdeb27716b7652d662c8d0953ae3d616c9cbae24c1c461c92585836b52",
53- "sha256:c348d663be126380630e3f50eda55aa16289c1d09431c5fa713e55735f7899a8",
54- "sha256:025c7d84cd6779b2d4833c8b88a839acda393d1b6118fa2b388270348527e739",
55- "sha256:ebf3c2077c5214036f32075b4611ed8d9b97395066ffa16d0808743999c4095d",
56- "sha256:0f9794fb6d8da30a4f3786c8b0e58c3797997ecb3cdd2fa41c99d53b72cb6ead",
57- "sha256:854475b282fa860b4740134ed656ed5466ae9c1d77d3add4907ad954c63b485a",
58- "sha256:2f29f878c6da6ce0ed2bc3130f76d491f8c8d53c5ab126e875df5df9ff0664db",
59- "sha256:b86e9e4ebcbcd7b84cf6296e2310acbfd3f699e5d70f449782403e65f55759d4",
60- "sha256:66794a8e5be715e2820605dbdd8b7a0eb1384ef1cbb0a02259d208cd3717fdd9"
```

```
3+ "LowerDir": "/mnt/nvme/990_PR0_4TB/docker/overlay2/yphztjyz5wx1k4m7s6v1necys/diff:
4+ /mnt/nvme/990_PR0_4TB/docker/overlay2/uqsqwgvy0k4mz7ok0b0eopvto/diff:
5- /mnt/nvme/990_PR0_4TB/docker/overlay2/tm4k9sqxxshmu6n5b33fq1up/diff:
6- /mnt/nvme/990_PR0_4TB/docker/overlay2/vny8kwjligdojx3kng5jwk9il/diff:
7- /mnt/nvme/990_PR0_4TB/docker/overlay2/t6wfnpfoo1yv7ocuhm9f7axum/diff:
8- /mnt/nvme/990_PR0_4TB/docker/overlay2/gzppnmckiq1pal7q836k9h4xz/diff:
9- /mnt/nvme/990_PR0_4TB/docker/overlay2/neqalsluljglzmx4a1iobjvn/diff:
10- /mnt/nvme/990_PR0_4TB/docker/overlay2/5zangfotyf9uyit7y5mt13dea/diff:
11- /mnt/nvme/990_PR0_4TB/docker/overlay2/d3qc25pr2etyfo7lzny761l8l/diff:
12- /mnt/nvme/990_PR0_4TB/docker/overlay2/3ymoas2002ia0qv4ix0wa0p5o/diff:
13- /mnt/nvme/990_PR0_4TB/docker/overlay2/roy4dl6iuzbxjfi7t4s6pkcqt/diff:
14- /mnt/nvme/990_PR0_4TB/docker/overlay2/l9li282rar9graqv55gw7o127/diff:
15- /mnt/nvme/990_PR0_4TB/docker/overlay2/a9yxjwwdq1dncqwt94z1lt8z/diff:
16- /mnt/nvme/990_PR0_4TB/docker/overlay2/6mqskqnye4b9g54daee4sr4l4/diff:
17- /mnt/nvme/990_PR0_4TB/docker/overlay2/f6y7lcyfkval9qb79cmp0r993/diff:
18- /mnt/nvme/990_PR0_4TB/docker/overlay2/kdbdeuudp5dshlc450ysuv5ns/diff:
19- /mnt/nvme/990_PR0_4TB/docker/overlay2/l27xq8i4yktlaa3j3n3r2aqc4/diff:
20- /mnt/nvme/990_PR0_4TB/docker/overlay2/m38ag01icnhxkzavz8vip7jmj/diff:
21- /mnt/nvme/990_PR0_4TB/docker/overlay2/
22- 72de23bd713fa6cc6f46c091d65a06653d619816c9ca1b2b9118d6f243dff50d/diff:
23- /mnt/nvme/990_PR0_4TB/docker/overlay2/
24- 34fc565b2b05ef51f167ff5ac9f7c5d8e12360bcd081d07b186448c75d0ab64/diff:
25+ /mnt/nvme/990_PR0_4TB/docker/overlay2/
26+ 4897143df74cda136c5323c6513f4d1dad8668f87a9ce00ca4b612b56d45a010/diff:
27+ /mnt/nvme/990_PR0_4TB/docker/overlay2/
28- 16e484f940327af75876efaf2035cc2adf54f795c3defea50389c5809d888f5e/diff",
29+ "MergedDir": "/mnt/nvme/990_PR0_4TB/docker/overlay2/margm6y6oo0t2c8wsy9m8m7rt/merged",
30+ "UpperDir": "/mnt/nvme/990_PR0_4TB/docker/overlay2/margm6y6oo0t2c8wsy9m8m7rt/diff",
31+ "WorkDir": "/mnt/nvme/990_PR0_4TB/docker/overlay2/margm6y6oo0t2c8wsy9m8m7rt/work"
32- },
33- "Name": "overlay2"
34- },
35- "RootFS": {
36- "Type": "layers",
37- "Layers": [
38- "sha256:418dcc7d85a63a6aa574439840f7a6fa6fd2321b3e2394568a317735e867d35",
39- "sha256:689195d1c9adf3fad82fbd3e2d279fa1bee001c73287e7e9ee2ed7a1881f2424",
40- "sha256:fddb0c9d0c8fa53f1466bdcc0baf00ad6ab1f3b5bcc5bdd152d3ce98f88ca16",
41- "sha256:00d0857b2a165e6b4dd62bd0d23f87e4e63a809a02b77333fe4351adadd52da7",
42- "sha256:754644eead3e70d8ddf4256935b64628cda167281e4425db8c29285fc83ed396",
43- "sha256:da963dab6af78b7e915a4c5f29966b14852df00bf99bec38c8e3a7400d83b9a9",
44- "sha256:f889c69583d2f7a7f1b01c0e15e023568cfd9a50e906eac9033ebb3d081dce99",
45- "sha256:36eb4331f20150d823f52aa204fe067201ba08b07ed57700400842bb1ade2045",
46- "sha256:2afdf1bef93cd49f4bc78cd6ddd5bb1bf9c2d86e060acbf117b86eb36ad5142c",
47- "sha256:eaed4d01f1befcd67ef089e2ef62f18cbaf7dca9898585ccb4b6968d379514771",
48- "sha256:bbdf35898477a1b8492e72abfe40835b46995b69fb0552bdaf77d1f500024f25",
49- "sha256:94eb5fa4282b81ab43f220baf41a84bca74b4c8389316792a4f20bba8fb7b10",
50- "sha256:5f70b1f18a086007016e948b04aed3b82103a36bea41755b6cddf10ace3c6ef",
51- "sha256:0bb3248bd9fe9a704ec8644eee9f4e6560c31d4b71faf6f7c10472efc478b18a",
52- "sha256:f01a81cdeb27716b7652d662c8d0953ae3d616c9cbae24c1c461c92585836b52",
53- "sha256:c348d663be126380630e3f50eda55aa16289c1d09431c5fa713e55735f7899a8",
54+ "sha256:95f2cc91b41092734313f52288d05f8e5841629b90dda1c9a3c2ac089d073bc",
55+ "sha256:566b5d0eff449f364aa3666558a6d01d345dbac78ceb9b7a55747ff4369b1651",
56+ "sha256:e018dabe57f73bde1d5e85e41834eade1b8900cfff6b10064fa10686a92fae205"
```

Dockerfile

- 이미지를 코드 형태로 정의하기 위한 일종의 표준 규격

Instruction	Description
ADD	Add local or remote files and directories.
ARG	Use build-time variables.
CMD	Specify default commands.
COPY	Copy files and directories.
ENTRYPOINT	Specify default executable.
ENV	Set environment variables.
EXPOSE	Describe which ports your application is listening on.
FROM	Create a new build stage from a base image.
HEALTHCHECK	Check a container's health on startup.
LABEL	Add metadata to an image.
MAINTAINER	Specify the author of an image.
ONBUILD	Specify instructions for when the image is used in a build.
RUN	Execute build commands.
SHELL	Set the default shell of an image.
STOPSIGNAL	Specify the system call signal for exiting a container.
USER	Set user and group ID.
VOLUME	Create volume mounts.
WORKDIR	Change working directory.

간단한 dockerfile

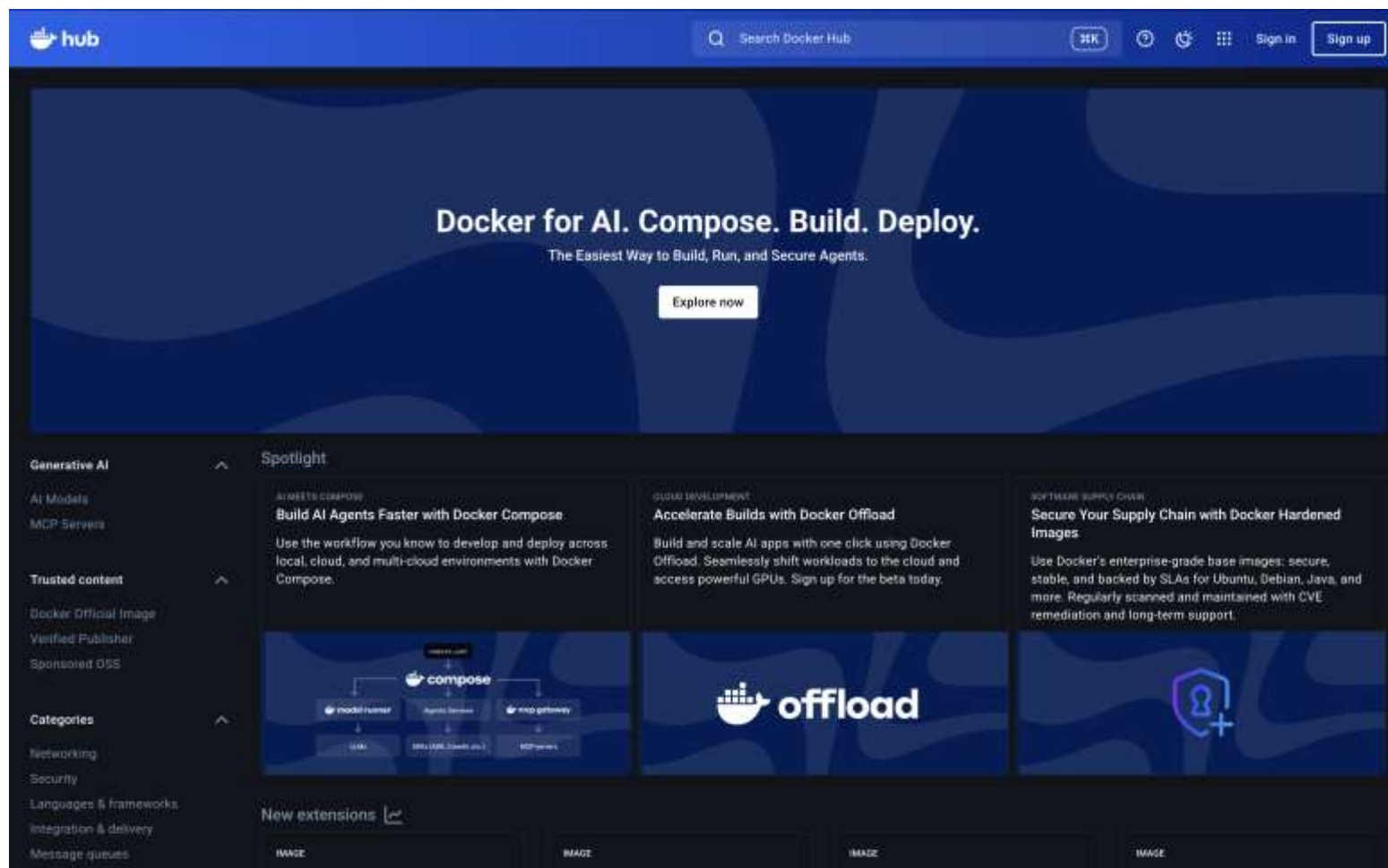
```
1  from flask import Flask
2  app = Flask(__name__)
3
4  @app.route("/")
5  def hello():
6      return "Hello World!"
```

```
1  # syntax=docker/dockerfile:1
2  FROM ubuntu:22.04
3
4  # install app dependencies
5  RUN apt-get update && apt-get install -y python3 python3-pip
6  RUN pip install flask==3.0.*
7
8  # install app
9  COPY hello.py /
10
11 # final configuration
12 ENV FLASK_APP=hello
13 EXPOSE 8000
14 CMD ["flask", "run", "--host", "0.0.0.0", "--port", "8000"]
```

- docker build -t test-build .
- docker run test-build

registry

- 이미지를 저장하고 관리하기 위한 서비스



Private registry



```
1 docker run -d -p 5000:5000 --restart always --name registry registry:3
```



```
1 docker pull ubuntu
2 docker tag ubuntu localhost:5000/ubuntu
3 docker push localhost:5000/ubuntu
```